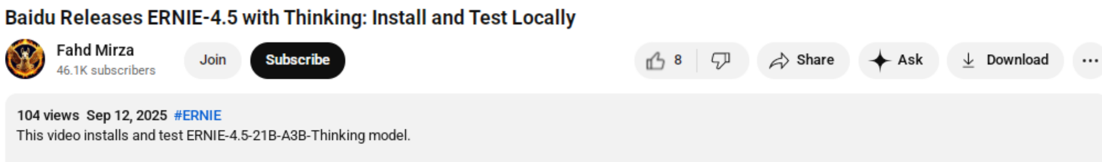
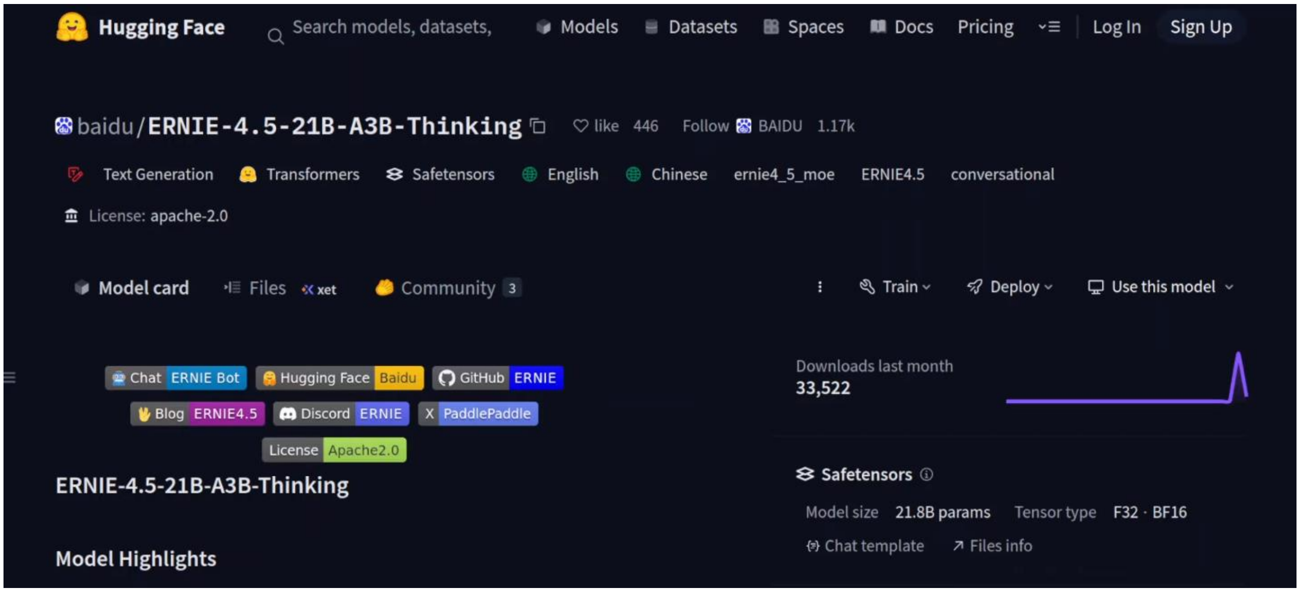
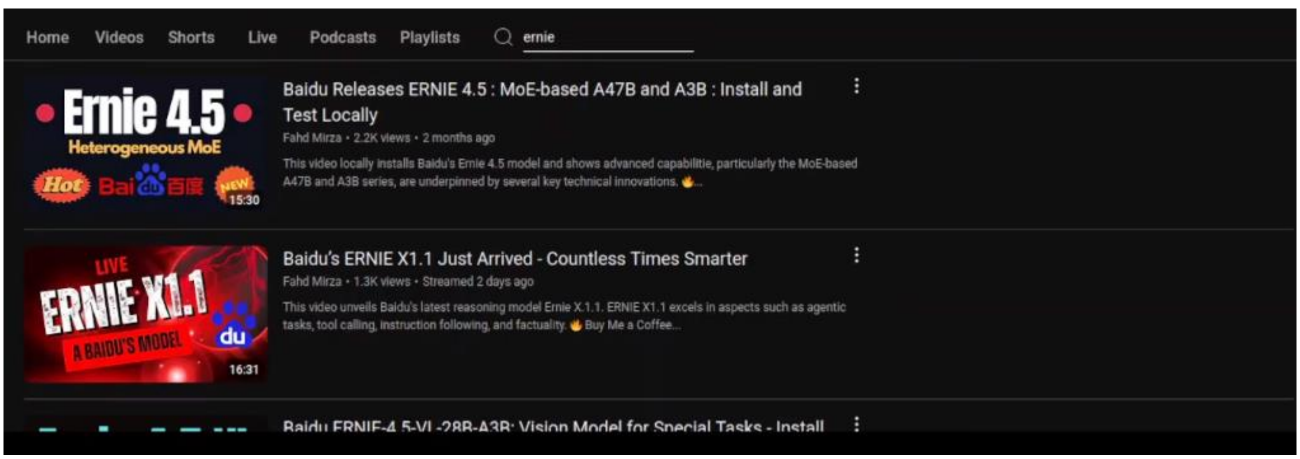
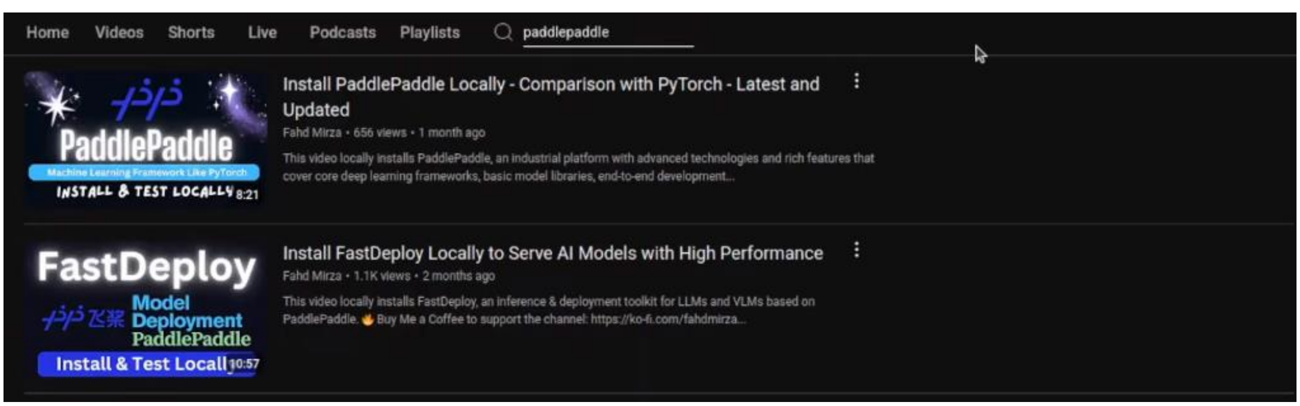
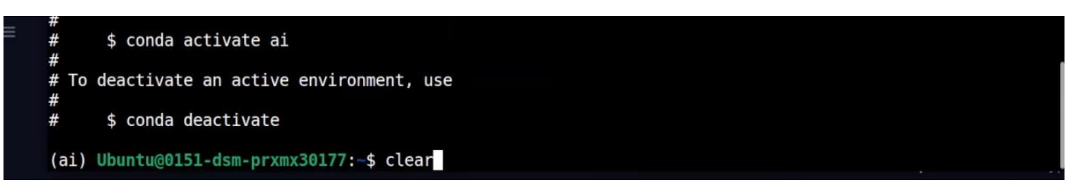


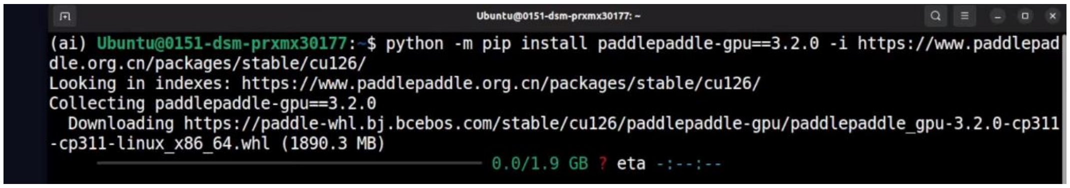


This is designed for complex reasoning tasks, supports 128K context length.



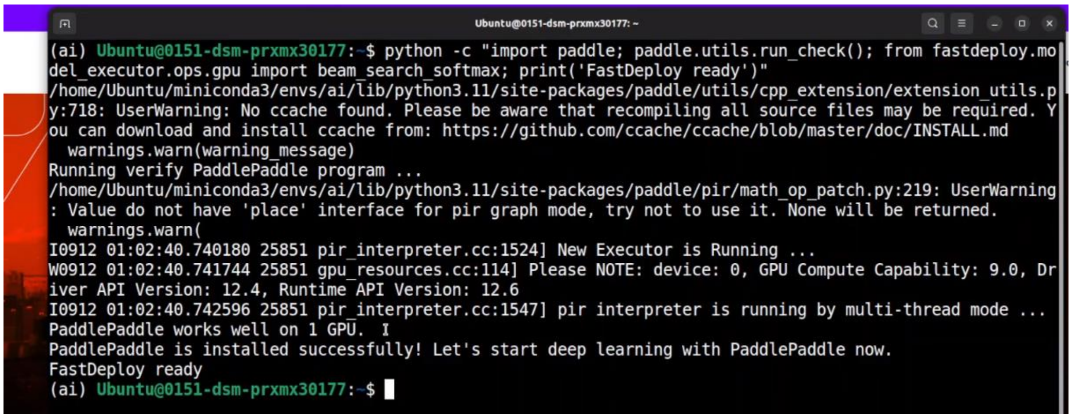






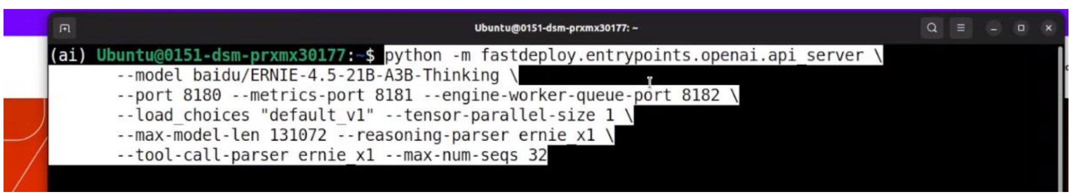
```
Ubuntu@0151-dsm-prxmx30177: ~
(ai) Ubuntu@0151-dsm-prxmx30177:~$ python -m pip install paddlepaddle-gpu==3.2.0 -i https://www.paddlepaddle.org.cn/packages/stable/cu126/
Looking in indexes: https://www.paddlepaddle.org.cn/packages/stable/cu126/
Collecting paddlepaddle-gpu==3.2.0
  Downloading https://paddle-whl.bj.bcebos.com/stable/cu126/paddlepaddle-gpu/paddlepaddle_gpu-3.2.0-cp311-cp311-linux_x86_64.whl (1890.3 MB)
                                        0.0/1.9 GB ? eta -:--:--
```

```
(ai) Ubuntu@0151-dsm-prxmx30177:~$ python -m pip install fastdeploy-gpu -i https://www.paddlepaddle.org.cn/packages/stable/fastdeploy-gpu-80_90/ --extra-index-url https://mirrors.tuna.tsinghua.edu.cn/pypi/web/simple
Looking in indexes: https://www.paddlepaddle.org.cn/packages/stable/fastdeploy-gpu-80_90/, https://mirrors.tuna.tsinghua.edu.cn/pypi/web/simple
Collecting fastdeploy-gpu
  Downloading https://paddle-whl.bj.bcebos.com/stable/fastdeploy-gpu-80_90/fastdeploy-gpu/fastdeploy_gpu-2.2.0-py3-none-any.whl (585.0 MB)
                                        0.0/585.0 MB ? eta -:--:--
```



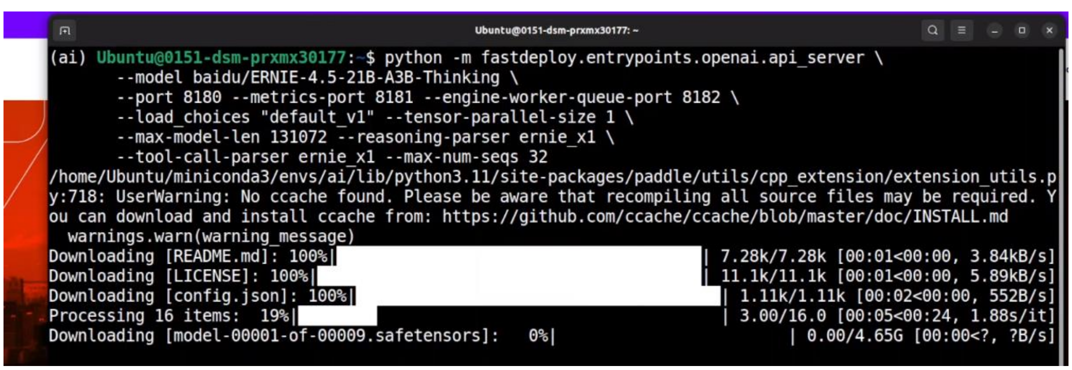
```
Ubuntu@0151-dsm-prxmx30177: ~
(ai) Ubuntu@0151-dsm-prxmx30177:~$ python -c "import paddle; paddle.utils.run_check(); from fastdeploy.model_executor.ops.gpu import beam_search_softmax; print('FastDeploy ready')"
/home/Ubuntu/miniconda3/envs/ai/lib/python3.11/site-packages/paddle/utils/cpp_extension/extension_utils.py:718: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
  warnings.warn(warning_message)
Running verify PaddlePaddle program ...
/home/Ubuntu/miniconda3/envs/ai/lib/python3.11/site-packages/paddle/pir/math_op_patch.py:219: UserWarning: Value do not have 'place' interface for pir graph mode, try not to use it. None will be returned.
  warnings.warn(
I0912 01:02:40.740180 25851 pir_interpreter.cc:1524] New Executor is Running ...
W0912 01:02:40.741744 25851 gpu_resources.cc:114] Please NOTE: device: 0, GPU Compute Capability: 9.0, Driver API Version: 12.4, Runtime API Version: 12.6
I0912 01:02:40.742596 25851 pir_interpreter.cc:1547] pir interpreter is running by multi-thread mode ...
PaddlePaddle works well on 1 GPU.
PaddlePaddle is installed successfully! Let's start deep learning with PaddlePaddle now.
FastDeploy ready
(ai) Ubuntu@0151-dsm-prxmx30177:~$
```

We are now ready to download and start doing inference with the LLM

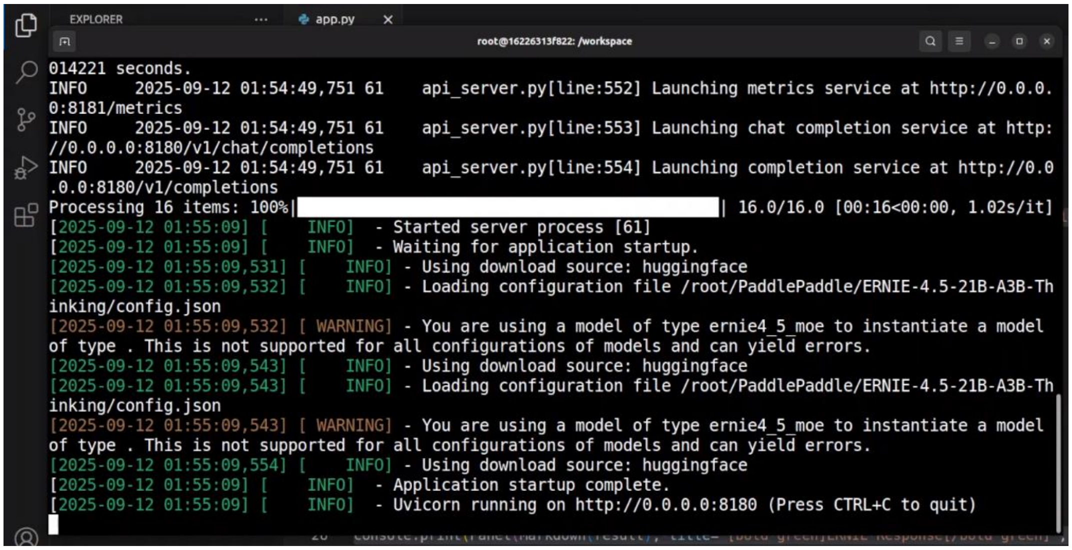


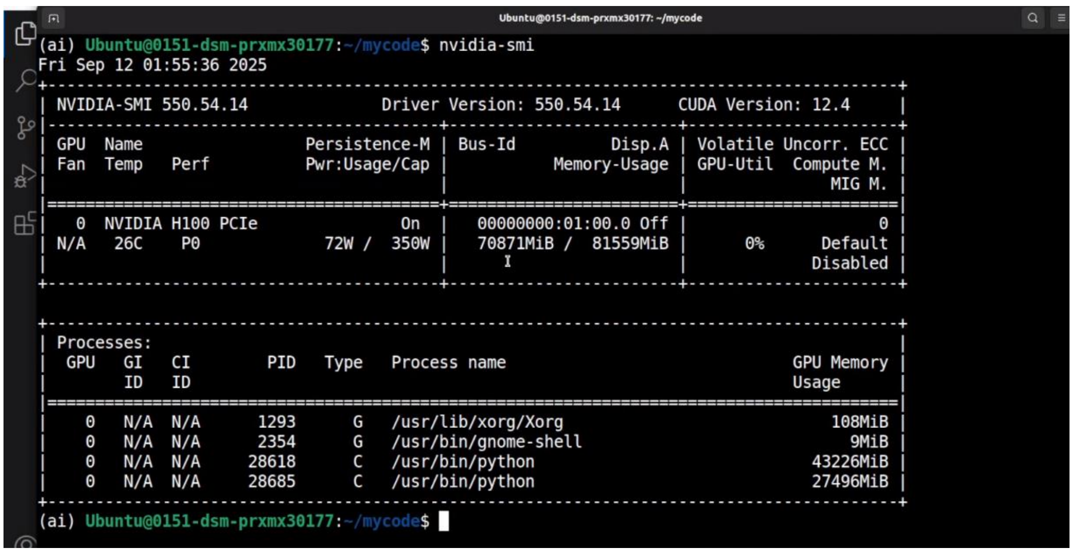
```
Ubuntu@0151-dsm-prxmx30177: ~
(ai) Ubuntu@0151-dsm-prxmx30177:~$ python -m fastdeploy.entrypoints.openai.api_server \
       --model baidu/ERNIE-4.5-21B-A3B-Thinking \
       --port 8180 --metrics-port 8181 --engine-worker-queue-port 8182 \
       --load_choices "default_v1" --tensor-parallel-size 1 \
       --max-model-len 131072 --reasoning-parser ernie_x1 \
       --tool-call-parser ernie_x1 --max-num-seqs 32
```

We will use **FastDeploy** engine to create an OpenAI API compatible endpoint

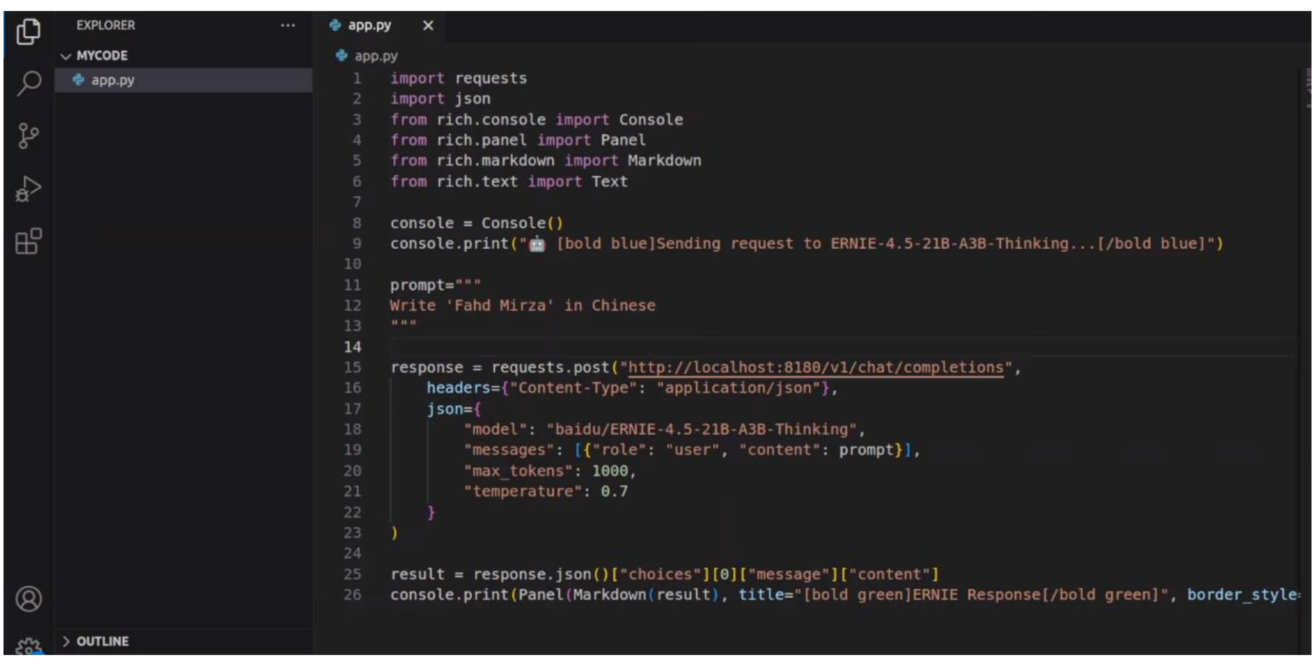


```
Ubuntu@0151-dsm-prxmx30177: ~
(ai) Ubuntu@0151-dsm-prxmx30177:~$ python -m fastdeploy.entrypoints.openai.api_server \
       --model baidu/ERNIE-4.5-21B-A3B-Thinking \
       --port 8180 --metrics-port 8181 --engine-worker-queue-port 8182 \
       --load_choices "default_v1" --tensor-parallel-size 1 \
       --max-model-len 131072 --reasoning-parser ernie_x1 \
       --tool-call-parser ernie_x1 --max-num-seqs 32
/home/Ubuntu/miniconda3/envs/ai/lib/python3.11/site-packages/paddle/utils/cpp_extension/extension_utils.py:718: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
  warnings.warn(warning_message)
Downloading [README.md]: 100%|          | 7.28k/7.28k [00:01<00:00, 3.84kB/s]
Downloading [LICENSE]: 100%|            | 11.1k/11.1k [00:01<00:00, 5.89kB/s]
Downloading [config.json]: 100%|        | 1.11k/1.11k [00:02<00:00, 552B/s]
Processing 16 items:  19%|             | 3.00/16.0 [00:05<00:24, 1.88s/it]
Downloading [model-00001-of-00009.safetensors]:   0%|          | 0.00/4.65G [00:00<?, ?B/s]
```



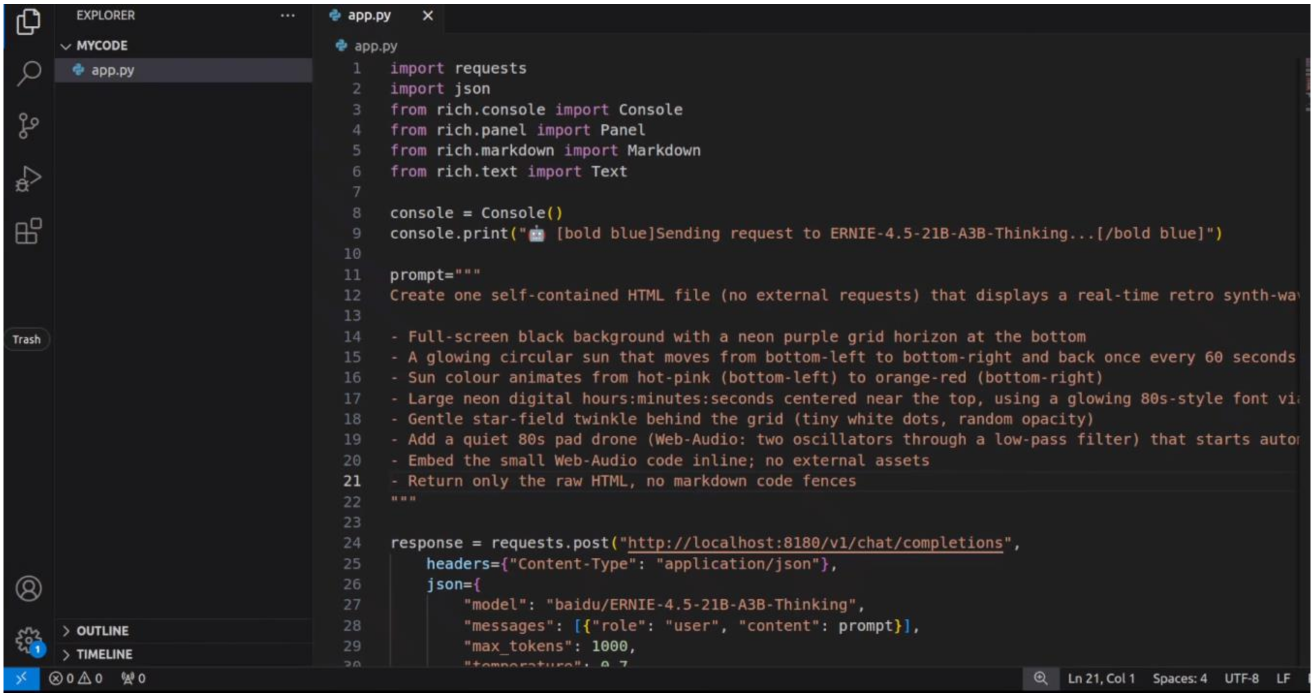


This is consuming close to 71GB of VRAM

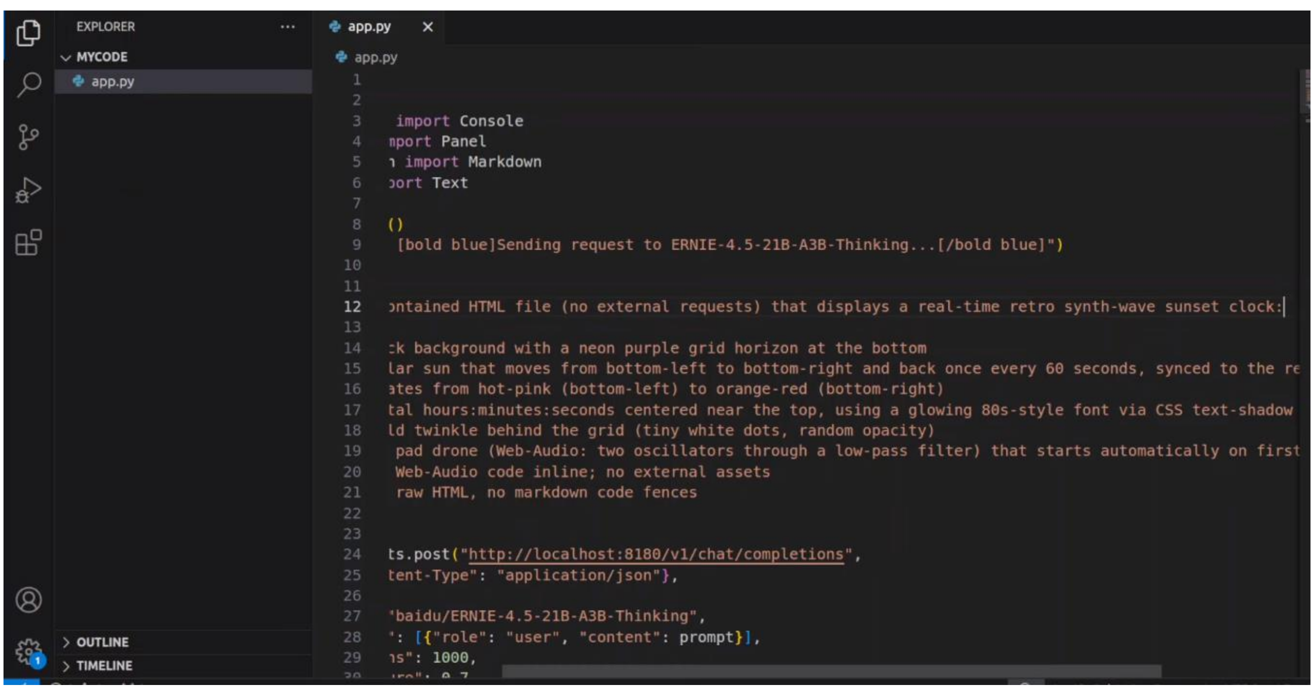


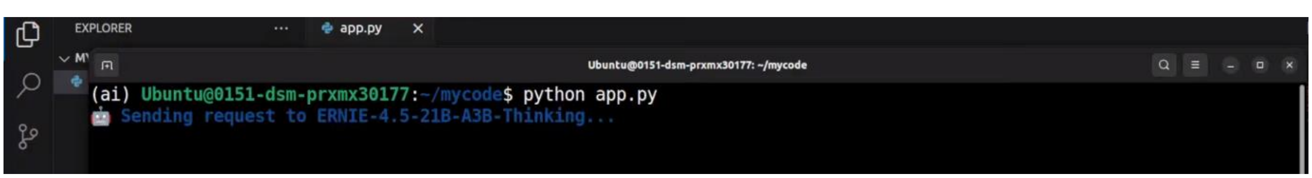
We will use this code to talk to the ERNIE LLM on its API endpoint.

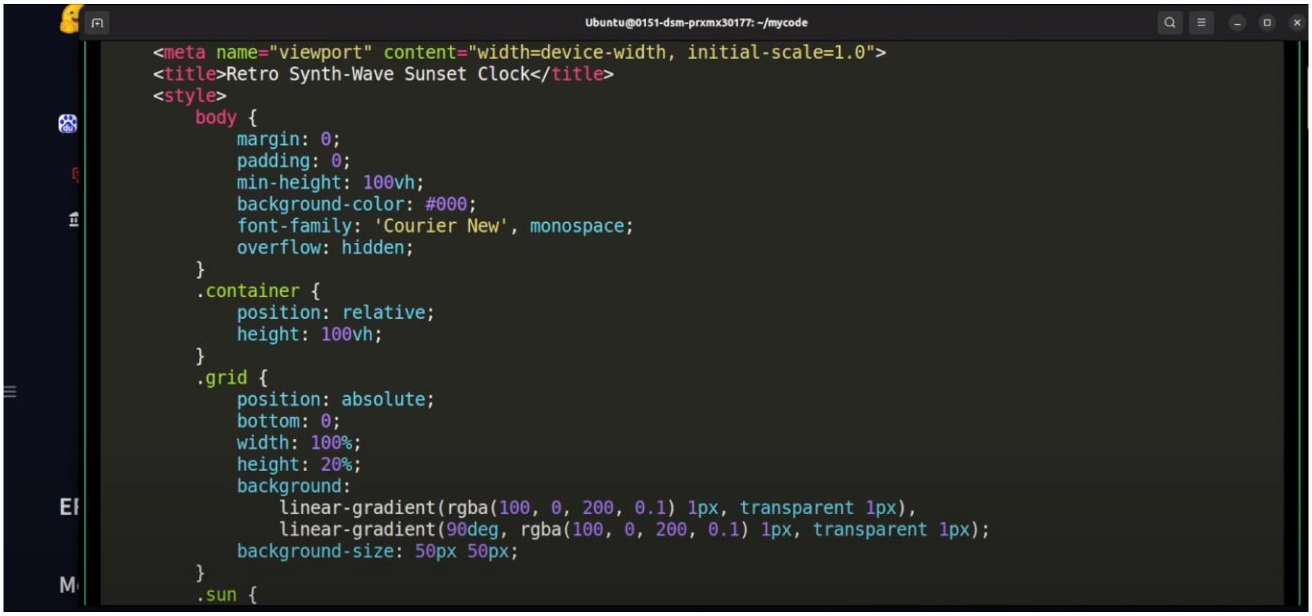
This is a bilingual model in English and Chinese as seen above

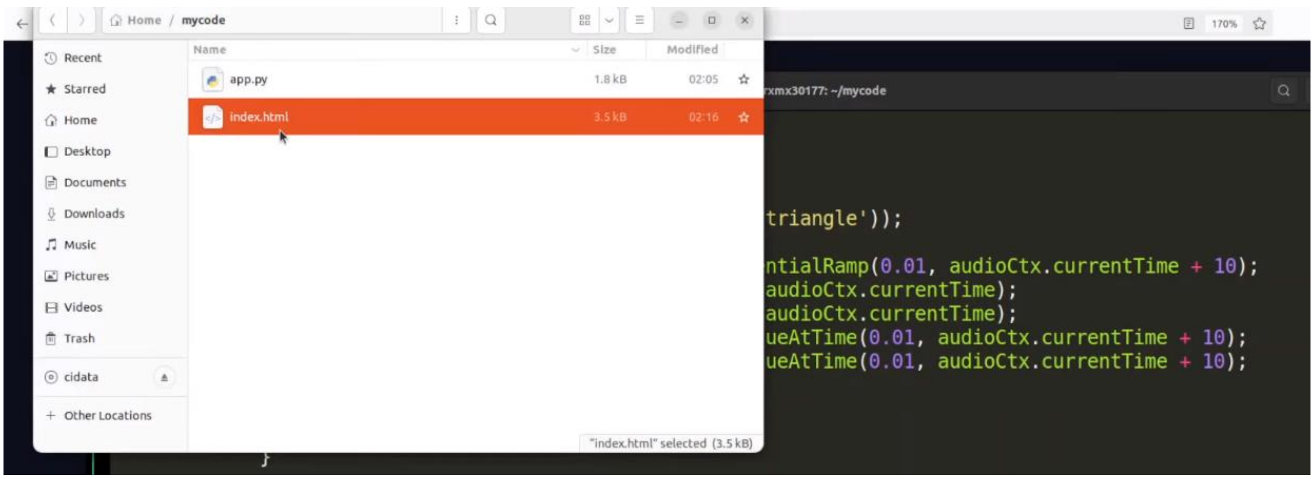


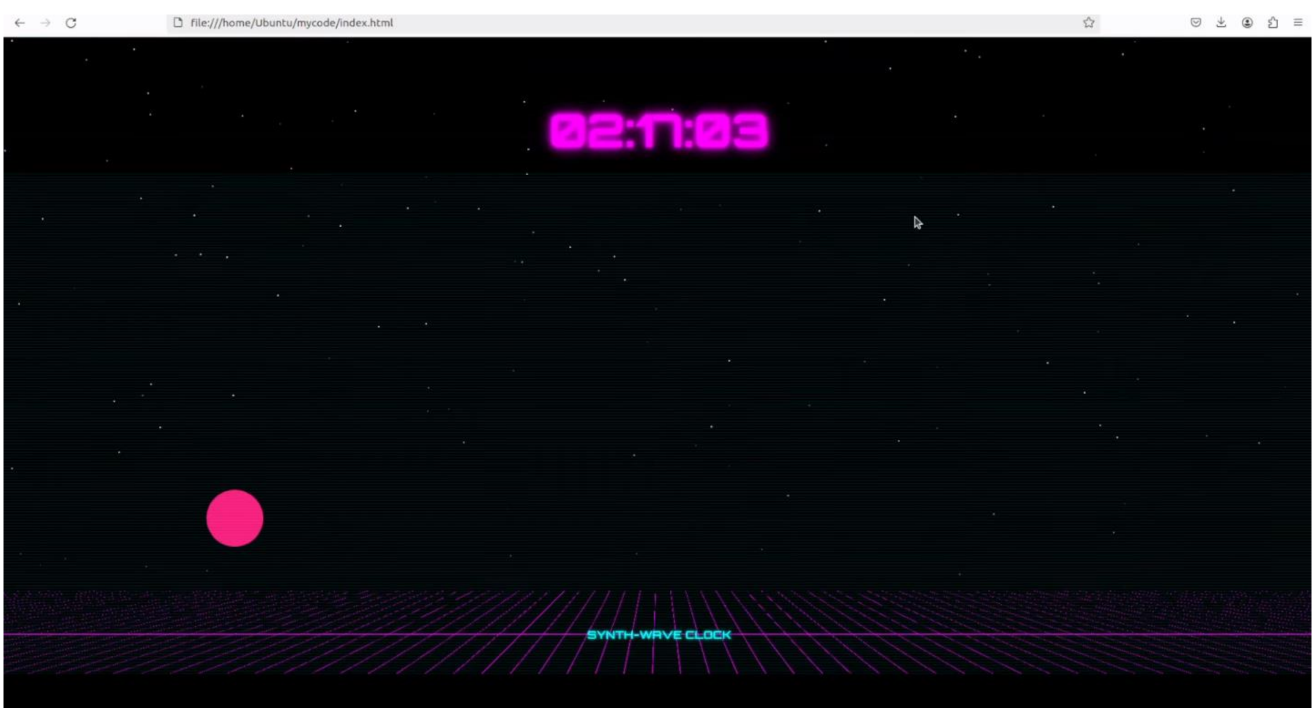
We can now do some inference on the API as above



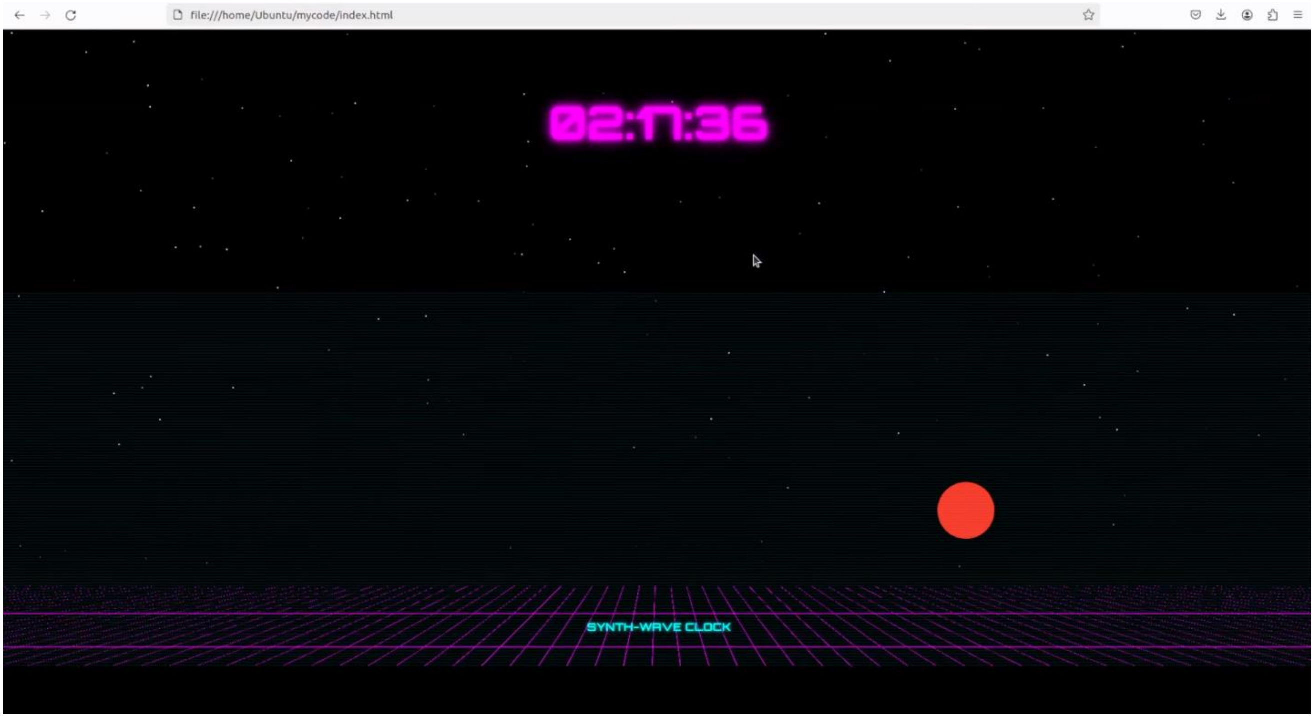


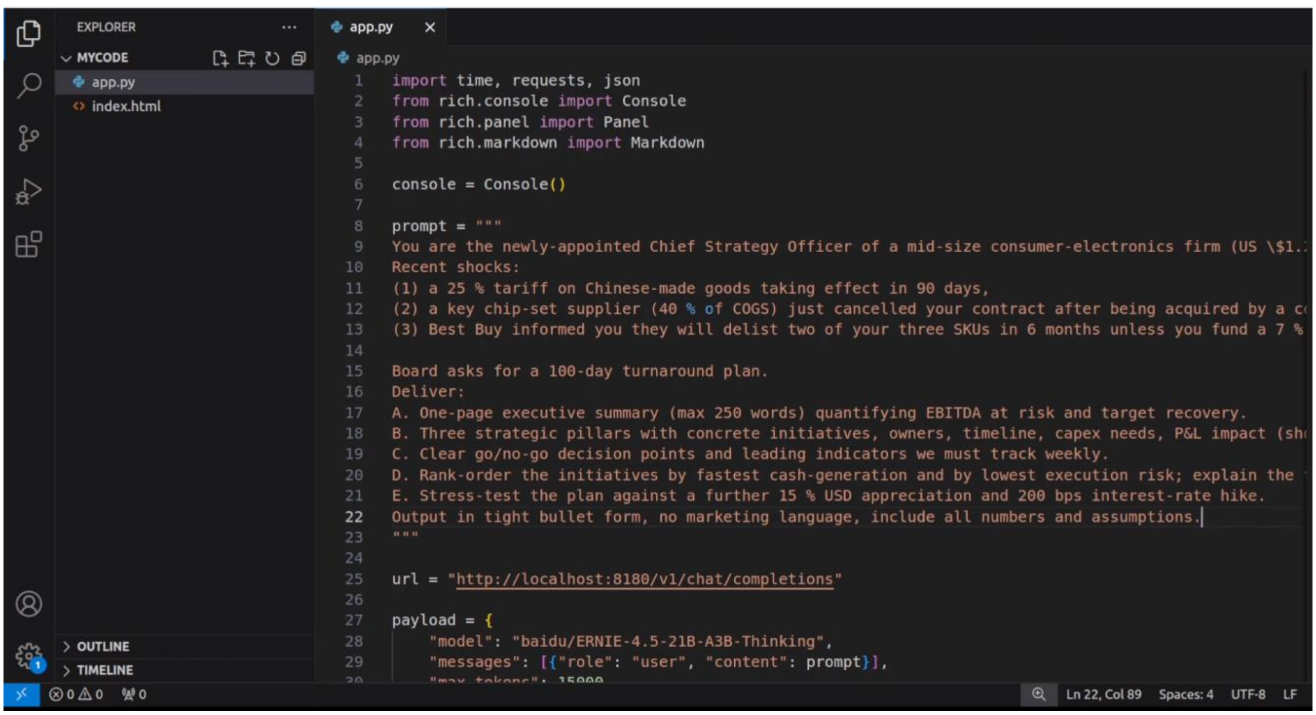




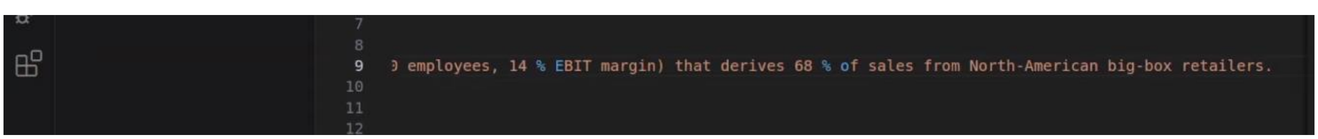


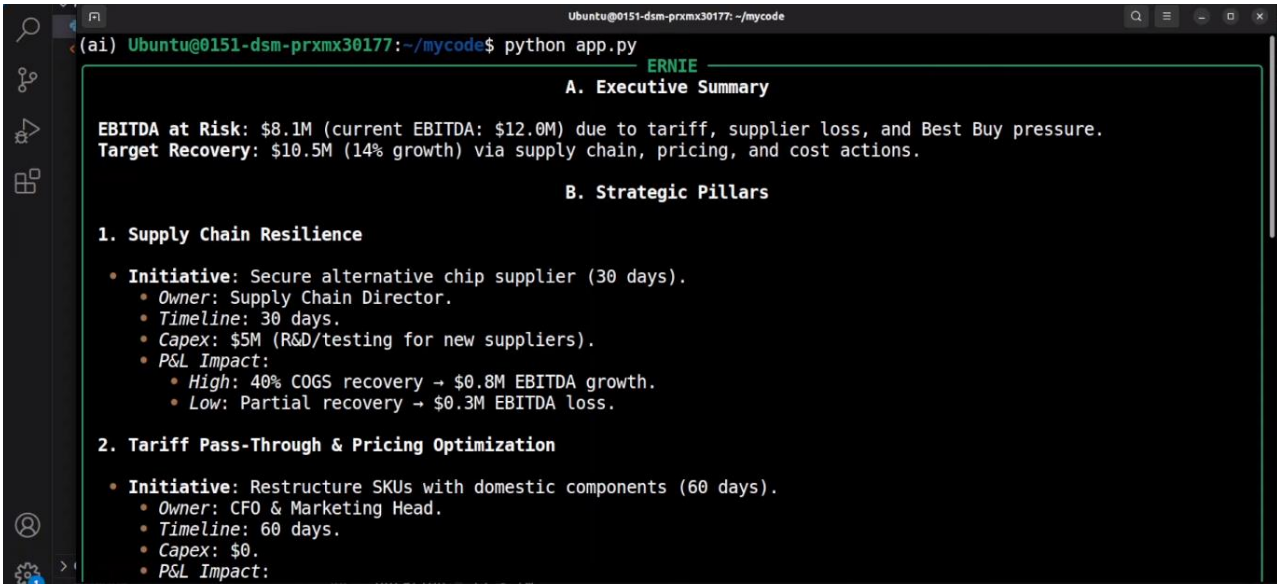
Looks pretty good!





Next, let us do a strategy stress test for the model





(ai) Ubuntu@0151-dsm-prxmx30177:~/mycode$ python app.py

ERNIE

A. Executive Summary

EBITDA at Risk: $8.1M (current EBITDA: $12.0M) due to tariff, supplier loss, and Best Buy pressure.
Target Recovery: $10.5M (14% growth) via supply chain, pricing, and cost actions.

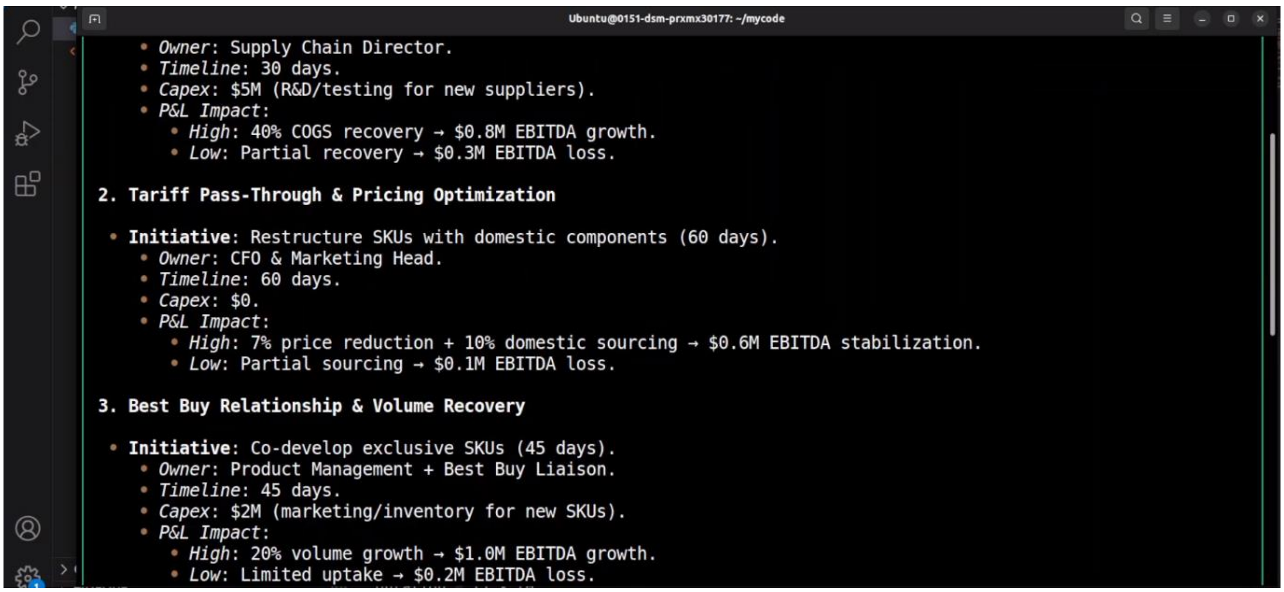
B. Strategic Pillars

1. Supply Chain Resilience

- **Initiative**: Secure alternative chip supplier (30 days).
  - *Owner*: Supply Chain Director.
  - *Timeline*: 30 days.
  - *Capex*: $5M (R&D/testing for new suppliers).
  - *P&L Impact*:
    - *High*: 40% COGS recovery → $0.8M EBITDA growth.
    - *Low*: Partial recovery → $0.3M EBITDA loss.

2. Tariff Pass-Through & Pricing Optimization

- **Initiative**: Restructure SKUs with domestic components (60 days).
  - *Owner*: CFO & Marketing Head.
  - *Timeline*: 60 days.
  - *Capex*: $0.
  - *P&L Impact*:



  - *Owner*: Supply Chain Director.
  - *Timeline*: 30 days.
  - *Capex*: $5M (R&D/testing for new suppliers).
  - *P&L Impact*:
    - *High*: 40% COGS recovery → $0.8M EBITDA growth.
    - *Low*: Partial recovery → $0.3M EBITDA loss.

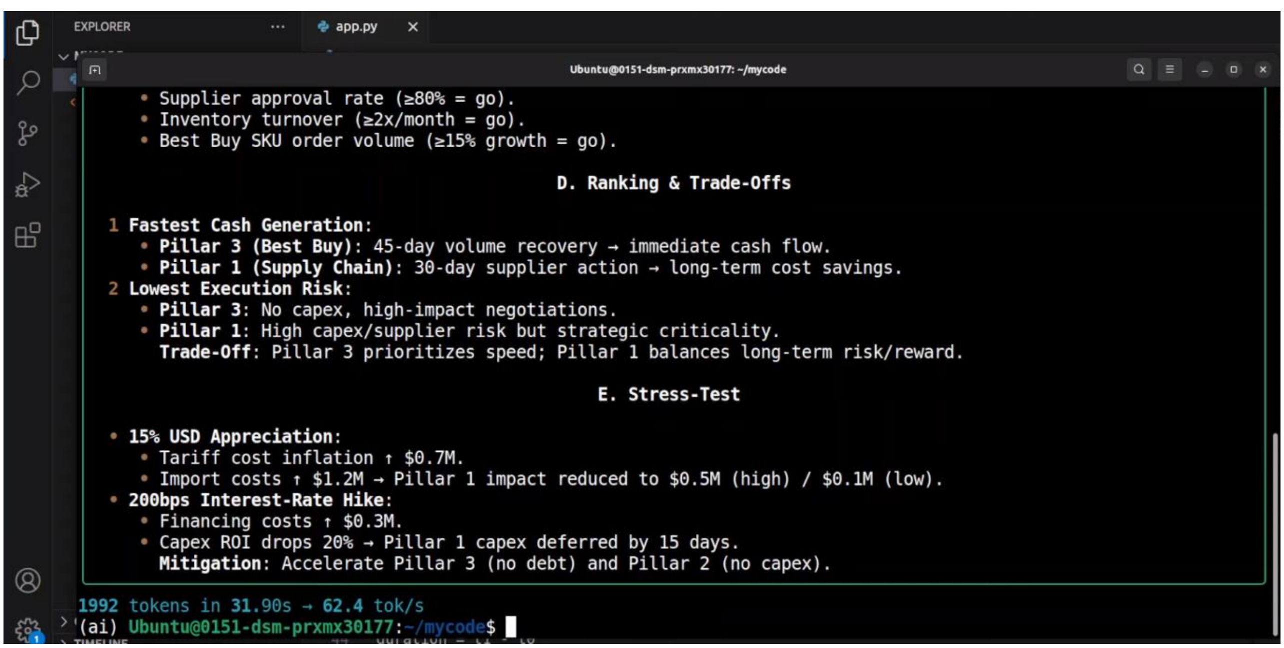
2. Tariff Pass-Through & Pricing Optimization

- **Initiative**: Restructure SKUs with domestic components (60 days).
  - *Owner*: CFO & Marketing Head.
  - *Timeline*: 60 days.
  - *Capex*: $0.
  - *P&L Impact*:

    - *High*: 7% price reduction + 10% domestic sourcing → $0.6M EBITDA stabilization.
    - *Low*: Partial sourcing → $0.1M EBITDA loss.

3. Best Buy Relationship & Volume Recovery

- **Initiative**: Co-develop exclusive SKUs (45 days).
  - *Owner*: Product Management + Best Buy Liaison.
  - *Timeline*: 45 days.
  - *Capex*: $2M (marketing/inventory for new SKUs).
  - *P&L Impact*:
    - *High*: 20% volume growth → $1.0M EBITDA growth.
    - *Low*: Limited uptake → $0.2M EBITDA loss.



EXPLORER ··· app.py

  - Supplier approval rate (≥80% = go).
  - Inventory turnover (≥2x/month = go).
  - Best Buy SKU order volume (≥15% growth = go).

D. Ranking & Trade-Offs

1 **Fastest Cash Generation**:
  - **Pillar 3 (Best Buy)**: 45-day volume recovery → immediate cash flow.
  - **Pillar 1 (Supply Chain)**: 30-day supplier action → long-term cost savings.

2 **Lowest Execution Risk**:
  - **Pillar 3**: No capex, high-impact negotiations.
  - **Pillar 1**: High capex/supplier risk but strategic criticality.
  Trade-Off: Pillar 3 prioritizes speed; Pillar 1 balances long-term risk/reward.

E. Stress-Test

- **15% USD Appreciation**:
  - Tariff cost inflation ↑ $0.7M.
  - Import costs ↑ $1.2M → Pillar 1 impact reduced to $0.5M (high) / $0.1M (low).
- **200bps Interest-Rate Hike**:
  - Financing costs ↑ $0.3M.
  - Capex ROI drops 20% → Pillar 1 capex deferred by 15 days.
  Mitigation: Accelerate Pillar 3 (no debt) and Pillar 2 (no capex).

1992 tokens in 31.90s → 62.4 tok/s
(ai) Ubuntu@0151-dsm-prxmx30177:~/mycode$

We can conclude that this model is good for creative writing, coding, step-by-step thinking tasks.